Fiscal Year:	FY 2021	**Task Last Updated:**	FY 06/08/2021
PI Name:	Doherty, Colleen Ph.D.		
Project Title:	Time and Space, Determining the Interactions Between the Circadian Clock and Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27695-7622	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	07/25/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/25/2022 per NSSC information (Ed., 8/9/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Perera, Imara Ph.D. (North Carolina State University)		
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Performance Goal Text:			

Task Description:	The circadian clock, an internal timekeeping mechanism, enables organisms to temporally organize their molecular and biochemical activities so that they are optimally timed with environmental conditions. The circadian clock contributes to overall fitness in plants and animals. Our preliminary results support a connection between plant gravity sensing and responses and the circadian clock, which has not been explored/reported previously. We propose to examine this interaction in both directions in Arabidopsis plants: investigating the effects of the circadian clock on gravity perception and responses and the system-level effects of simulated microgravity on the molecular clock. In early spaceflight studies, the impacts of microgravity on the circadian clock and the subsequent effects on physiology in humans and other organisms were recognized, as well as the potential for profound implications for crew performance. In mammalian studies, there is evidence that microgravity and recovery from microgravity affect core components of the mammalian circadian clock. Yet a full examination of the molecular interplay between gravity and the circadian clock has not been reported in any organism. As with crew-performance, a disrupted circadian clock has a negative impact on plant growth and fitness. In plants, almost all aspects of growth, physiology, and environmental responses are modulated by the circadian clock. Therefore, understanding the interplay between changes in gravity, the circadian clock, and circadian-regulated output will have important implications for plant growth in future space missions. The experiments outlined here will identify not only the effects of changes in gravity levels on the core clock components but will also examine the genome-wide downstream effects on transcriptional timing. We observed that the time of day a gravitropic stimulus is provided affects the magnitude of the response. Furthermore, this response is altered in a circadian mutant. Based on these findings we hypothesize that, like other environmental stimuli, the response to gravity is gated by the circadian clock. We also examined transcriptomic datasets in GeneLab and found that core clock components are differentially expressed in response to gravity in multiple studies in Arabidopsis. This suggests that the circadian clock may be sensitive to changes in gravity. To examine how the interaction between the circadian clock and gravity is affected under simulated microgravity we will utilize the random positioning machine to evaluate the following hypotheses. Hypothesis 1: Gravity sensor(s) is/are controlled by the circadian clock. Therefore, the time of day the plants are transferred to microgravity will affect their response and the response to gravity will be altered in plants with mutations in the circadian clock. Objective: Determine the effects of the time of day and the circadian clock on gravitropic responses. Hypothesis 2: Gravity signals modulate the Arabidopsis circadian clock and circadian-regulated outputs. Therefore, clock components and/or clock outputs will be altered under simulated microgravity. Objective: Investigate how simulated microgravity affects the circadian clock and circadian-regulated outputs. This proposal focuses on the interaction between gravity and the circadian clock in a plant system, yet many principles of circadian circuits span across multiple organisms. Transcriptomic data will be generated and deposited into the GeneLab database. The system-level understanding of gravity perception and responses and the intersection with the circadian clock acquired here will contribute towards adapting plants to spaceflight and may also improve our understanding of the impacts of space travel on other organisms. Training of new generation US Space Biologists will focus on data analysis and integration of data sets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has two investigation tracks which have an impact on life on Earth. The first aspect is to understand how the time of day and the circadian clock affect the response to gravity in plants. The circadian clock controls many aspects of growth and development in plants and does this through daily regulation of transcription, translation, and other biochemical activities. Since most of the molecular components of the plant show a daily rhythm in their activities, the pathways and responses that depend on them also vary in efficiency throughout the day. We are using this underlying variation in biochemistry and physiology to uncover the pathways that plants use to sense and respond to changes in gravity. Understanding plant responses to gravity can have significant impacts on life on Earth as plant root growth, controlled by gravity, is an important factor in the ability of plants to access water. Our second investigation track focuses on the inverse connection--does gravity affect the circadian clock? The circadian clock of plants and animals enables organisms to optimize their response to the environment coordinating their internal molecular activities with externally recurring events such as sunrise and sunset or the changing of seasons. The internal circadian clock has been shown to enhance fitness in many species, and an altered clock can affect performance (e.g., yield in plants, response time in humans, and resistance to disease in both plants and animals). This molecular clock has developed in organisms within the confines of Earth with an ~24 h diel cycle and under ~1g of gravity. We are investigating if changes in gravity affect the functioning of the molecular clock, either through altering the oscillator itself or the output regulated by the circadian oscillator. If the clock is impacted by gravity, this can be used to understand the clock mechanisms. Insights into how the clock works and how it is connected to the downstream activities it regulates can be used to enhance the performance of crops, particularly as changing climate conditions allow us to grow crops in new environments. Two critical changes in plant growth are the poleward movement of cultivated areas and the expansion of growing seasons throughout the year. Both of these aspects involve components that are regulated by the circadian clock. Therefore, understanding how the clock functions can provide us with strategies for enhancing plant adaptability to new environmental conditions.
Task Progress:	Task progress during this reporting period was impacted by Covid-related limitations on personnel occupancy in the laboratory, travel restrictions, and supply availability. The primary progress of this period is in the completion of the investigation into the effects of time of day and the circadian clock on gravitropic responses. This part of the project has been compiled as a manuscript and submitted. Additionally, the methods section of this part was prepared as a methods manuscript in the compiled edition of Gravitropic Methods. Much less progress has been made on the second focus of this project examining how gravity impacts the circadian clock. This part was supposed to be the main focus of this past project period. We have completed some preliminary trials to optimize the plant growth on the RPM (random positioning machine) apparatus developed by Jacob Torres. We anticipate initiating these experiments at Microgravity Simulation Support Facility (MSSF) when the university travel restrictions are lifted July 6 2021.
Bibliography Type:	Description: (Last Updated: 05/25/2026)

Books/Book Chapters	Tolsma JS, Torres JJ, Richards JT, Perera IY, Doherty CJ. "Evaluating the effects of the circadian clock and time of day on plant gravitropic responses." in "Plant Gravitropism. Ch 19. Methods Molecular Biology, vol. 2368." Ed. E.B. Blancaflor. New York: Humana, e2021. in press as of June 2021. , Jun-2021